



**Politecnico
di Torino**

Politecnico di Torino

Nuclear fission plants

Laboratory on flow patterns and two-phase pressure drops

Master degree in Sustainable nuclear energy

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Chapter 1

Introduction

During the laboratory session, different experimental tests were carried out, with different water and air flow rates, in order to study two-phase flow. The main parameters investigated are the pressure drop and the void fraction. The aims of the laboratory can be sum up as:

- (i) **Simulation and Experimental Data Acquisition:** Simulate two-phase air-water flow through a vertical pipe to observe and identify different flow pattern types while simultaneously measuring the experimental parameters.
- (i) **Evaluation of Experimental Fluid-Dynamic Parameters:** Process the acquired measurements to evaluate the average void fraction (α) within the test section and determine the experimental pressure difference (Δp) by applying fluid statics principles to the differential pressure transducer readings.
- (i) **Theoretical Modeling and Performance Comparison:** Predict void fraction and pressure drops using various theoretical models and correlations (such as Homogeneous, Zivi, Chisholm, CISE, Drift Flux, and Friedel) and compare these results with experimental findings through the use of flow pattern maps (Hewitt-Roberts and Taitel-Dukler).

Chapter 2

Experimental Procedures and Data Acquisition

2.1 Experimental Facility and Instrumentation

The experimental facility for upward flow is sketched in Figure 2.1. The main components, arranged according to the fluid flow path, are described below:

1. **Water Storage and Recirculation:** Water is drawn by pumps from a dedicated tank and is subsequently recirculated to the same tank after passing through the test section.
2. **Thermoresistance (PT100):** A PT100 thermoresistance is installed upstream of the diaphragms to measure the water temperature with high accuracy.
3. **Bourdon Pressure Transducers:** Relative pressure transducers of the Bourdon type are utilized to monitor pressure levels downstream of the pump and upstream of the diaphragms, as well as at the inlet of each rotameter.
4. **Water Flow Rate Measurement (Diaphragms):** A set of three diaphragms is mounted on parallel pipes to measure the water mass flow rate across different ranges: 0.12–0.32 kg/s, 0.3–1.1 kg/s, and 0.5–3.0 kg/s. The differential pressure across these devices is measured by a Rosemount transducer, and the flow rate is calculated using specific calibration curves and correlations.
5. **Air Flow Rate Measurement (Rotameters):** Air flow is regulated via three rotameters in parallel, operating across varying ranges: 0.2–1.2, 0.45–4.2, and 1.5–17.0 Nm³/h. These instruments utilize a tapered vertical pipe where the position of an internal float (or bullet) indicates the flow rate, reaching

an equilibrium position where its weight is balanced by the fluid's drag and friction forces.

6. **Air-Water Mixer:** A porous bronze device located at the inlet of the vertical pipe ensures the uniform mixing of the two phases.
7. **Test Section:** The mixture flows through a vertical plexiglass pipe with an inner diameter of 26 mm. This section is equipped with two pneumatic fast-closing valves that close simultaneously to trap the two-phase mixture, allowing for the experimental evaluation of the average void fraction through a draining and weighing method.
8. **Phase Separator:** At the exit of the test section, the two-phase mixture enters a vertical cylindrical separator; here, the air is vented from the top while the water is recirculated back to the main tank.

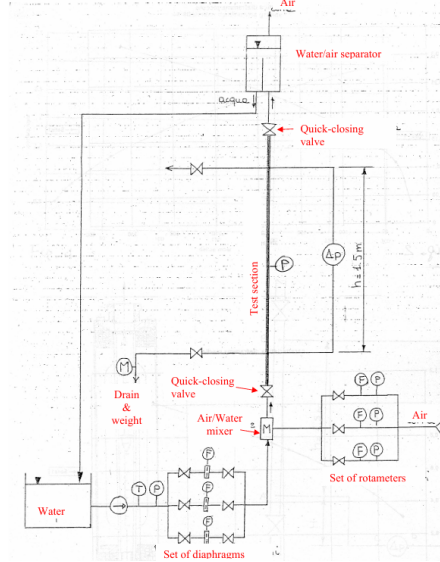


Figure 2.1: Sketch of upward flow facility

2.2 Experimental data and fluid-dynamic parameters

The experimental data measured are summarized in Table 2.1. Due to a technical issue during data acquisition, the Δp_{transd} value for the 16th record was not recorded. This test run has been excluded from all pressure drop calculations.

Table 2.1: Experimental data: Two-phase water-air flux parameters.

Test #	Diaph size	Q_{air} (Nm^3/h)	P_{rot} (bar)	ΔP_{diafr} (mbar)	P_{rel} (bar)	ΔP type	ΔP_{pipe} (mbar)	T ($^{\circ}C$)	M_{water} (g)	Flow Pattern
1	S	0.25	6.0	76.35	0.29	PD-PC (+)	71.17	24.82	370	Slug
2	S	0.73	5.9	76.11	0.27	PD-PC (+)	87.18	25.12	283	Slug
3	S	2.94	5.0	75.52	0.27	PD-PC (+)	91.77	24.92	216	Slug
4	S	8.43	5.4	75.62	0.26	PD-PC (+)	97.23	24.62	125	Slug/Churn
5	S	15.12	4.2	75.34	0.27	PD-PC (+)	96.27	24.33	59	Churn/Annular
6	S	0.25	6.0	462.50	0.31	PD-PC (+)	18.28	20.81	620	Bubble
7	S	0.72	5.9	462.50	0.32	PD-PC (+)	27.65	20.13	530	Bubble/Slug
8	S	2.88	5.2	462.50	0.32	PD-PC (+)	36.3	19.25	328	Slug
9	S	8.31	5.4	244.62	0.28	PD-PC (+)	19.43	18.37	175	Slug
10	S	15.26	4.2	243.23	0.29	PD-PC (+)	72.83	18.27	112	Slug/Churn
11	S	2.8	5.2	244.77	0.28	PD-PC (+)	79.91	17.88	236	Slug
12	M	0.29	6.0	85.37	0.32	PD-PC (+)	24.77	16.81	578	Bubble
13	M	0.71	5.9	85.41	0.31	PD-PC (+)	36.83	16.81	503	Slug
14	M	2.9	5.2	85.32	0.31	PD-PC (+)	52.12	16.81	323	Slug
15	M	8.36	5.4	84.45	0.33	PD-PC (+)	27.86	17.01	221	Slug
16	M	15.37	4.2	83.23	0.36	/	/	17.32	151	Slug/Churn
17	M	0.28	6.0	215	0.34	PD-PC (+)	12.01	17.69	643	Bubble
18	M	2.89	5.2	213.3	0.34	PD-PC (+)	17.41	17.88	387	Slug
19	M	8.3	5.4	209.1	0.39	PC-PD (-)	22.68	17.88	283	Slug/Churn
20	M	15.09	4.3	204.36	0.44	PC-PD (-)	57.76	18.08	206	Slug/Churn

Once the raw experimental data were gathered, an analytical procedure was implemented to calculate the fluid-dynamic parameters of interest. Thermophysical properties of water and air (density ρ , dynamic viscosity μ , and surface tension σ) were evaluated at the actual operational temperatures and pressures using the CoolProp library. The test section geometry was defined by an inner diameter $D = 0.026$ m and a measurement length $h = 1.5$ m.

2.2.1 Evaluation of the Water flow rate

Each diaphragm was calibrated in laboratory in order to get characteristic curves, described by the following analytical relationship:

$$W = \alpha_{MQ} \frac{\pi d^2}{4} \sqrt{2\rho\Delta p} + B$$

where the constants are showed in Table 2.2

Table 2.2: Diaphragms parameters

Diaphragm size	$d(m)$	α_{MQ}	B
small	0.0084	0.650729	0.01415274
medium	0.0153	0.728193	0.00375718

2.2.2 Evaluation of Air flow rate

The rotameters are calibrated at normal condition, and so are the result provided. Assuming air temperature at 20 °C, the actual mass flow rate can be evaluated as follow:

$$W_e = 1.204 \sqrt{\frac{p_{e,rel} + 1.013}{1.013}} \cdot Q_s$$

where $p_{e,rel}$ is the relative pressure measured by the manometer at the rotameter inlet and Q_s is the value in Nm^3/h given by the rotameter.

2.2.3 Evaluation of the Void Fraction

The void fraction (α) represents the volumetric fraction of the test section occupied by the gaseous phase. The experimental evaluation of the average void fraction was performed using the quick-closing valve technique. By simultaneously closing the pneumatic valves at the inlet and outlet, a representative volume of the two-phase mixture was trapped. The liquid mass (M_l) was determined by weighing the

drained mass (M_{dr}) and adding the residual mass (M_{res}) trapped in the lower part of the pipe, estimated at 0.116 kg. Given the total liquid mass required to fill the pipe ($M_{l0} = 1.014$ kg), the void fraction is calculated as:

$$\alpha_{exp} = 1 - \frac{M_{dr} + M_{res}}{M_{l0}}$$

2.2.4 Evaluation of the Pressure Drops

The total pressure difference between the lower tap (p_A) and the upper tap (p_B) is derived from the readings of the Rosemount differential pressure transducer (Δp_{transd}). A critical operational constraint of this specific transducer is its ability to measure only positive pressure differences.

Consequently, the plastic lines connecting the pressure taps to the device must be properly configured so that the difference between the "High" and "Low" chambers remains positive under the specific flow conditions of each test. The choice of connection scheme depends on whether the friction pressure drop (Δp_{frict}) is greater or smaller than the hydrostatic term $(\rho - \bar{\rho})gh$. By applying the principles of fluid statics to the measurement lines, the actual pressure change along the test section is evaluated as follows:

- Lower tap to "High" chamber ($p_C - p_D > 0$): In this case, the transducer measures $\Delta p_{transd} = p_C - p_D$, and the total pressure drop is calculated as:

$$p_A - p_B = \Delta p_{transd} + \rho_w gh$$

- Higher tap to "High" chamber ($p_D - p_C > 0$): The transducer measures $\Delta p_{transd} = p_D - p_C$, and the relationship becomes:

$$p_A - p_B = \rho_w gh - \Delta p_{transd}$$

In these equations, $h = 1.5$ m represents the vertical distance between the taps, and ρ_w is the density of the water inside the connection pipes.

Table 2.3: Correlation result for total pressure drop [Pa].

Test	W_{air} [kg/s]	W_{water} [kg/s]	α_{exp} [-]	Δp_{exp} [Pa]
1	0.0002	0.1549	0.5207	7598
2	0.0006	0.1546	0.6065	5997
3	0.0024	0.1341	0.6726	5538
4	0.0071	0.1542	0.7623	4992
5	0.0115	0.1539	0.8274	5088
6	0.0002	0.3606	0.2742	12887
7	0.0006	0.3607	0.3629	11950
8	0.0024	0.3607	0.5621	11085
9	0.0070	0.2662	0.7130	12772
10	0.0116	0.2655	0.7751	7432
11	0.0023	0.2663	0.6529	6724
12	0.0003	0.5566	0.3156	12238
13	0.0006	0.5568	0.3895	11032
14	0.0024	0.5565	0.5671	9503
15	0.0070	0.5536	0.6677	11929
16	0.0117	0.5496	0.7367	—
17	0.0002	0.8811	0.2515	13514
18	0.0024	0.8776	0.5039	12974
19	0.0070	0.8689	0.6065	16983
20	0.0116	0.8591	0.6824	20491

Chapter 3

Theoretical Modeling and Correlations

To initiate the theoretical analysis, the thermophysical properties of water and air—including density (ρ), dynamic viscosity (μ), and surface tension (σ)—are first determined using the CoolProp library based on the experimental temperature and pressure recorded for each test. However, the predictive performance of the subsequent correlations relies primarily on the evaluation of global fluid-dynamic parameters. The thermodynamic mass quality (x) is calculated to define the gas mass fraction as:

$$x = \frac{W_{air}}{W_{air} + W_{water}} \quad (3.1)$$

The total mass flux (G) is determined to characterize the flow intensity per unit of cross-sectional area (A_{cross}):

$$G = \frac{W_{air} + W_{water}}{A_{cross}} \quad (3.2)$$

Furthermore, the superficial velocities for the gas (j_g) and liquid (j_l) phases are evaluated, along with the total superficial velocity (j_{tot}):

$$j_g = \frac{W_{air}}{A_{cross} \cdot \rho_g}, \quad j_l = \frac{W_{water}}{A_{cross} \cdot \rho_l}, \quad j_{tot} = j_g + j_l \quad (3.3)$$

These parameters serve as the fundamental coordinates for flow pattern identification and are essential for calculating the volumetric interactions and momentum exchange between the phases.

3.1 Evaluation of the Void Fraction

To predict the volumetric distribution of the phases, several theoretical models and correlations were implemented. All of them except the Drift Flux Model, evaluate the void fraction with the following equation, with different formulation for the slip ratio S

$$\alpha = \frac{1}{1 + \left(\frac{1-x}{x}\right) \frac{\rho_g}{\rho_l} \cdot S} \quad (3.4)$$

- **Homogeneous Model:** This model assumes that the gas and liquid phases move at the same velocity, implying a slip ratio

$$S = 1$$

This model is particularly accurate for low void fraction values.

- **Zivi Correlation:** Zivi proposed a model where the slip ratio is determined solely by the density ratio, seeking to minimize the kinetic energy of the flow. The slip ratio is defined as:

$$S = \left(\frac{\rho_l}{\rho_g}\right)^{1/3}$$

This model is particularly accurate for high void fraction values.

- **Chisholm Correlation:** This correlation introduces a dependence of the slip ratio on the mass quality x , allowing for a more dynamic representation of phase interactions:

$$S = \sqrt{1 - x \left(1 - \frac{\rho_l}{\rho_g}\right)}$$

- **CISE Correlation:** Recognized for its higher accuracy, the CISE model evaluates the slip ratio through empirical parameters (E_1, E_2) and dimensionless Reynolds (Re) and Weber (We) numbers. The slip ratio is calculated as:

$$S = 1 + E_1 \left(\frac{y}{1 + yE_2} - yE_2 \right)^{0.5}$$

where the intermediate fluid-dynamic and dimensionless parameters are defined by the following relations:

$$\beta = \frac{\rho_l x}{\rho_l x + \rho_g (1 - x)}, \quad y = \frac{\beta}{1 - \beta}$$

$$Re = \frac{GD}{\mu_l}, \quad We = \frac{G^2 D}{\sigma \rho_l}$$

$$E_1 = 1.578 Re^{-0.19} \left(\frac{\rho_l}{\rho_g} \right)^{0.22}, \quad E_2 = 0.0273 We Re^{-0.51} \left(\frac{\rho_l}{\rho_g} \right)^{-0.08}$$

In these equations, β represents the volumetric quality, while E_1 and E_2 account for the combined effects of mass flux, pipe geometry, and surface tension on the phase distribution.

- **Drift Flux Model:** This mechanistic model accounts for the non-uniform distribution of phases and the buoyancy-induced velocity of the gas:

$$\alpha = \frac{j_g}{C_0 j + u_{gj}} \quad (3.5)$$

The distribution parameter was set to $C_0 = 1.13$ and the drift velocity u_{gj} was calculated using the bubbly flow correlation:

$$u_{gj} = 1.41 \left[\frac{\sigma g (\rho_l - \rho_g)}{\rho_l^2} \right]^{0.25}$$

3.2 Evaluation of the Pressure Drop

The total pressure change along the vertical test section is evaluated by considering the primary contributions of gravitational (elevation) and frictional effects. Neglecting the acceleration pressure change—associated with local variations in density and void fraction—the total pressure difference between the measurement taps is defined as:

$$p_A - p_B = \bar{\rho} g h + \Delta p_{fric} \quad (3.6)$$

where $\bar{\rho}$ is the average mixture density.

The calculation of the friction component (Δp_{fric}) requires the determination of a friction factor (f), which is selected based on the flow regime identified by the Reynolds number (Re). Following the implementation in the analytical script, the Darcy correlation is utilized for laminar flow, while the Blasius correlation is applied for turbulent conditions:

- **Laminar flow** ($Re < 2000$): $f = \frac{64}{Re}$
- **Turbulent flow** ($Re \geq 2000$): $f = 0.316 Re^{-0.25}$

Table 3.1: Correlation results for void fraction.

Test	α_{hom}	α_{Zivi}	α_{Chisholm}	α_{CISE}	$\alpha_{\text{drift flux}}$
1	0.4776	0.0957	0.3980	0.3016	0.3100
2	0.7294	0.2370	0.5842	0.4806	0.5431
3	0.9211	0.5735	0.7679	0.6995	0.7665
4	0.9681	0.7769	0.8471	0.8130	0.8380
5	0.9799	0.8488	0.8775	0.8559	0.8551
6	0.2790	0.0431	0.2474	0.1925	0.2030
7	0.5234	0.1136	0.4314	0.3495	0.4053
8	0.8064	0.3270	0.6477	0.5827	0.6745
9	0.9446	0.6633	0.8026	0.7715	0.8175
10	0.9656	0.7649	0.8418	0.8267	0.8427
11	0.8498	0.3951	0.6877	0.6178	0.7087
12	0.2241	0.0326	0.2029	0.1616	0.1723
13	0.4143	0.0760	0.3513	0.2849	0.3291
14	0.7326	0.2418	0.5868	0.5258	0.6163
15	0.8882	0.4817	0.7278	0.7070	0.7692
16	0.9285	0.6046	0.7782	0.7775	0.8102
17	0.1479	0.0199	0.1381	0.1143	0.1185
18	0.6287	0.1656	0.5081	0.4561	0.5320
19	0.8279	0.3634	0.6671	0.6539	0.7173
20	0.8860	0.4828	0.7254	0.7350	0.7730

In this study, two distinct modeling approaches were employed to evaluate the dissipative terms:

- **Homogeneous Model:** This approach treats the air-water mixture as a single-phase pseudo-fluid with averaged thermophysical properties. The thermodynamic mass quality (x) is used to derive the homogeneous density (ρ_h) and a weighted viscosity ($\mu_h = x\mu_g + (1 - x)\mu_l$), which are then used to compute the homogeneous Reynolds number. The frictional pressure drop is expressed as:

$$\Delta p_{fric} = f_{hom} \frac{h}{D} \frac{G^2}{2\rho_h} \quad \text{where} \quad \rho_h = \left(\frac{x}{\rho_g} + \frac{1-x}{\rho_l} \right)^{-1} \quad (3.7)$$

- **Friedel model:** This is a separated flow approach that evaluates the friction pressure drop by applying a two-phase multiplier (ϕ_{l0}^2) to the pressure drop calculated assuming the total mass flux flows as a liquid (Δp_{l0}). The friction pressure drop is calculated as:

$$\Delta p_{fric} = \Delta p_{l0} \cdot \phi_{l0}^2 \quad \Delta p_{l0} = f_{l0} \frac{h}{D} \frac{G^2}{2\rho_l}$$

The Friedel multiplier is defined as:

$$\phi_{l0}^2 = E + \frac{3.24 \cdot F \cdot H}{Fr_h^{0.045} \cdot We_h^{0.035}}$$

where the required dimensionless numbers and the mixture density are:

$$Fr_h = \frac{G^2}{gD\rho_h^2}, \quad We_h = \frac{G^2D}{\sigma\rho_h}, \quad \rho_h = \left(\frac{x}{\rho_g} + \frac{1-x}{\rho_l} \right)^{-1}$$

The empirical parameters E , F , and H are determined by the following relationships:

$$E = (1-x)^2 + x^2 \frac{\rho_l}{\rho_g} \frac{f_{g0}}{f_{l0}}, \quad F = x^{0.78} (1-x)^{0.224}$$

$$H = \left(\frac{\rho_l}{\rho_g} \right)^{0.91} \left(\frac{\mu_g}{\mu_l} \right)^{0.19} \left(1 - \frac{\mu_g}{\mu_l} \right)^{0.7}$$

In these equations, f_{l0} and f_{g0} are the friction factors calculated considering the total flow rate as either completely liquid or completely gaseous, respectively.

The calculation of the elevation pressure drop component ($\bar{\rho}gh$) depends on the mixture density, which is a function of the void fraction:

$$\bar{\rho} = \alpha\rho_g + (1-\alpha)\rho_l \quad (3.8)$$

This contribution can be calculated using each of the models described in Section 3.1.

Table 3.2: Correlation results for total pressure drop [Pa].

Test	Hom–Hom	Friedel–Zivi	Friedel–Chisholm	Friedel–CISE	Friedel–drift flux
1	7831	13580	9152	10564	10442
2	4286	11773	6686	8203	7289
3	1984	7403	4557	5559	4576
4	3069	5965	4937	5436	5071
5	4462	5970	5550	5865	5877
6	11100	14889	11893	12697	12543
7	7780	14397	9736	10937	10119
8	4767	12828	8124	9078	7732
9	4806	9354	7310	7766	7091
10	6956	9580	8451	8673	8438
11	3683	11081	6788	7813	6479
12	12436	15774	13275	13881	13724
13	9980	15853	11814	12789	12140
14	6932	15799	10736	11631	10303
15	8761	16330	12718	13023	12111
16	12048	17512	14965	14974	14494
17	14603	17239	15505	15854	15793
18	10207	19675	14650	15413	14300
19	12617	22512	18057	18250	17321
20	16622	24808	21249	21108	20551

Chapter 4

Comparison and Performance Analysis

This chapter presents a comparison between the experimental data and the theoretical predictions calculated for void fraction, total pressure drops, and flow patterns. Comparison diagrams are provided where the theoretical results are plotted against experimental values. These diagrams include a diagonal line representing perfect agreement, alongside dashed lines indicating a $\pm 20\%$ deviation band to highlight the reliability of each predictive approach.

Figure 4.1 illustrates the performance of the Homogeneous, Zivi, Chisholm, and Drift Flux models against the measured void fraction. It is observed that while the Homogeneous model tends to overestimate the void fraction, the Chisholm and Drift Flux correlations generally provide a better fit within the $\pm 20\%$ error bands.

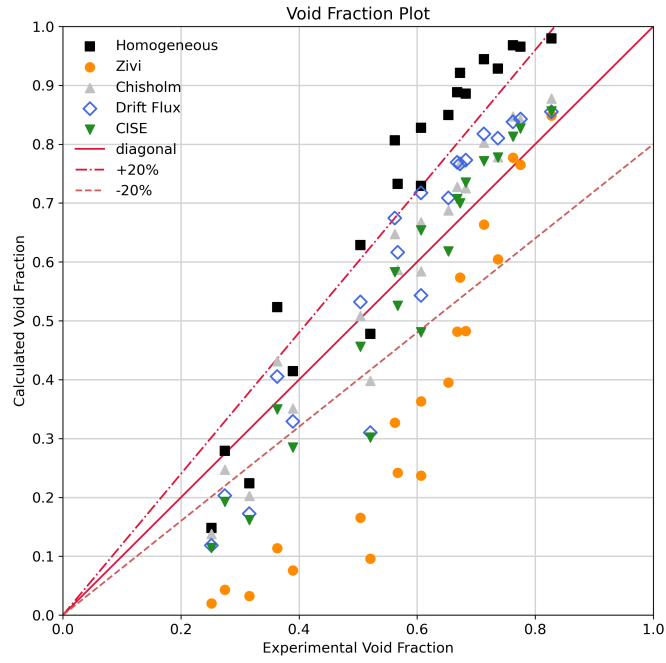


Figure 4.1: Comparison of experimental and calculated void fraction (α).

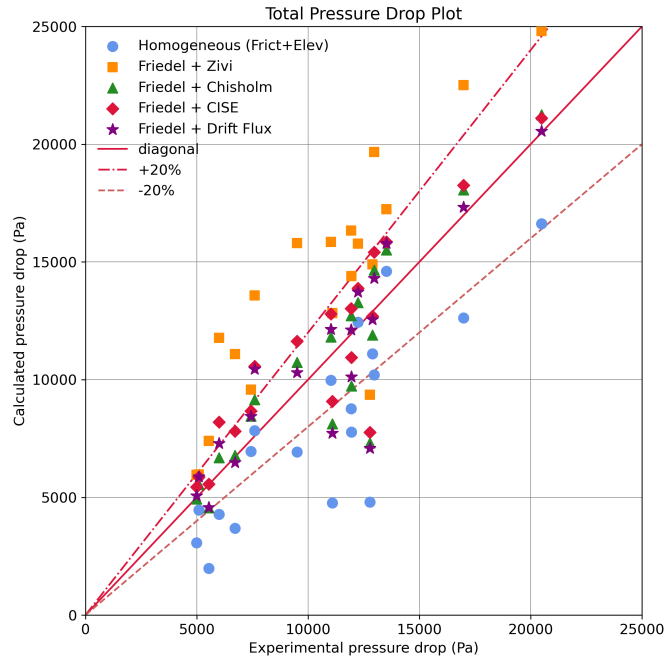


Figure 4.2: Comparison between experimental and predicted total pressure drops.

Figure 4.2 reports the total pressure drops calculated using several combinations of friction (Homogeneous or Friedel) and elevation (Zivi, Chisholm, CISE, or Drift Flux) models. The diagram highlights that the Homogeneous approach often underestimates the total pressure drop, whereas the combinations involving the Friedel friction multiplier show higher predictive accuracy.

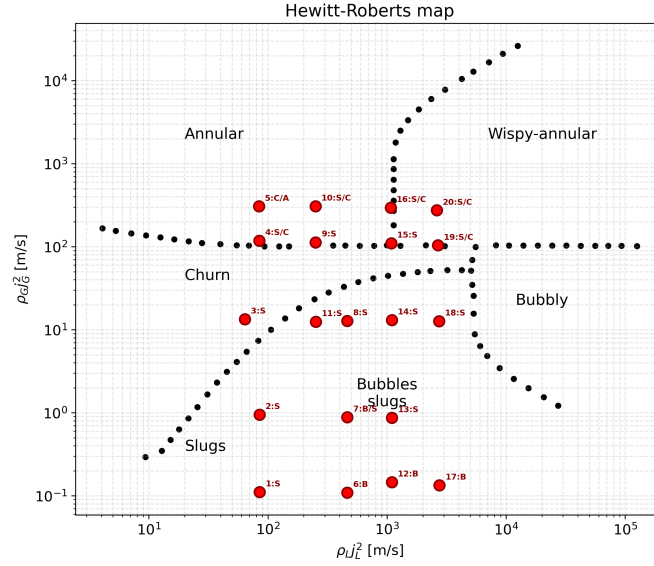


Figure 4.3: Experimental data points located on the Hewitt-Roberts flow pattern map.

The Hewitt-Roberts map (Figure 4.3) displays the experimental points according to the superficial momentum flux of each phase ($\rho_L j_L^2$ vs. $\rho_G j_G^2$). This map is used to verify if the observed flow regimes (Bubbly, Slug, Churn, or Annular) match the theoretically predicted regions.

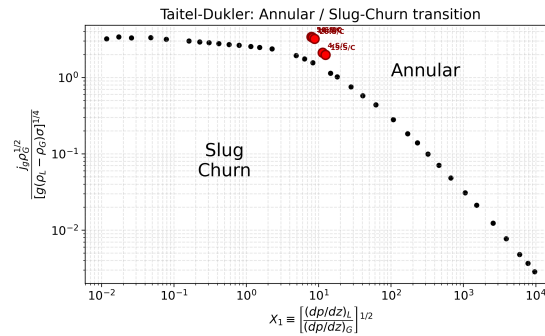


Figure 4.4: Taitel-Dukler transition map: Annular vs. Slug-Churn flow.

Figure 4.4 focuses on the transition between the Annular and Slug-Churn regimes, utilizing the Martinelli parameter X as a coordinate to evaluate the stability of the liquid film.

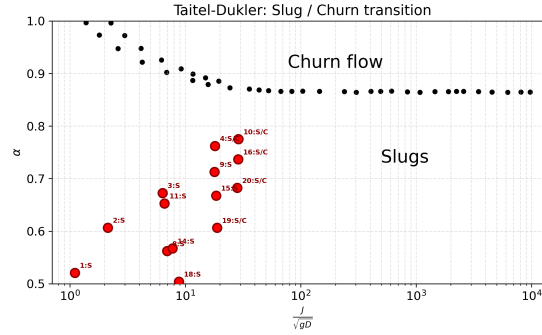


Figure 4.5: Taitel-Dukler transition map: Slug vs. Churn flow.

The Slug/Churn transition is detailed in Figure 4.5, where the experimental void fraction is plotted against the dimensionless superficial velocity $j/\sqrt{gD}$ to assess the breakdown of the slug regime.

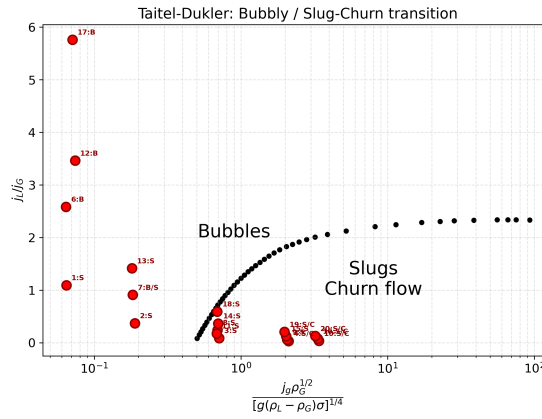


Figure 4.6: Taitel-Dukler transition map: Bubbly vs. Slug-Churn flow.

Finally, Figure 4.6 represents the transition between Bubbly and Slug/Churn flow, determined by the ratio of superficial velocities j_L/j_G and the gas phase properties.